

Philosophies of Funding

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Successful scientific practice encompasses broader and more varied modes of investigation than can be captured by focusing on hypothesis-driven research. We examine the emphases that major US and UK funding agencies place on particular modes of research practice and suggest that funding agency guidelines should be informed by a more dynamic and multidimensional account of scientific practice.

Philosophy is often regarded as abstract and irrelevant to practical concerns. Yet, when it comes to decisions about funding science, philosophical conceptions of good scientific methodology have major implications for research policy. A number of scientists and philosophers have argued that the best science is hypothesis driven and that science's pivotal activity is to test hypotheses. From this perspective, descriptive, exploratory, and inductive methodologies, although sometimes necessary, are fundamentally preparatory. They are seen as valuable primarily to the extent that they fuel hypothesis-driven science. Despite this common view, a great deal of evidence indicates that it may be misleading. Examples from physics, chemistry, and biology suggest that accounts of scientific activity should involve not only the proposal and testing of hypotheses, but also the exploration of phenomena, the development of technology and techniques, and the generation of questions that can be addressed with inductive, model-building approaches (Glass and Hall, 2008; Kell and Oliver, 2004; Caldin, 2002).

This wider viewpoint sees the primary output of science as its ability to intervene in and understand the natural world. Producing and testing hypotheses are important but insufficient strategies for achieving these aims. We maintain that an adequate theoretical account of how science works will need to subordinate the generation and testing of hypotheses to a far more inclusive range of practices, and that one of

the most crucial arenas in which this widening of perspective must occur is in the funding of science. Although we believe that inspiration, effort, community structure, and intellectual rewards are of deep importance to scientific progress, from a practical viewpoint in today's research environment, almost all scientific inquiry requires official funding of some sort. Because funding agencies use philosophical conceptions of good scientific methodology to guide funding decisions, philosophy affects the progress of science. It is therefore important that funding philosophies reflect the complete body of strategies used to achieve scientific ends.

Here, we consider how our claims about the nature of scientific practice relate—and should relate—to policy statements about the practice of research and to funding agency guidelines for grant applications. To do this, we focus on statements from the National Institutes of Health (NIH) and the National Science Foundation (NSF) in the US and the Biological and Biomedical Sciences Research Council (BBSRC) in the UK. We examine how these statements depict both hypothesis-driven and more exploratory, question-driven, or technology-oriented modes of investigation. We conclude that developing a more complete representation of the iterative, interdisciplinary, and multidimensional relationships between various modes of scientific investigation could improve funding agency guidelines.

A Broad Account of Scientific Practice

It is standard for scientists, philosophers, and other commentators on scientific practice to claim that hypothesis testing lies at the very heart of scientific practice and that it is, in fact, the singular activity that demarcates it from nonscientific activities (Popper, 1963). From a historical point of view, observers of science know this is problematic. The concept of the hypothesis and the popularity of hypothesis-driven modes of research have been in flux throughout the history of science (Laudan, 1981). The familiar representation of science as hypothesis driven is typically based on situations in which very specific questions can be addressed within tightly bounded spheres of inquiry. Because such testing is upheld as an idealization of scientific practice, broader inquiries may often be framed post-hoc as narrower hypothesis testing (Glass, 2006). Other modes of practice, however, are at least as prevalent in less artificially narrowed contexts of inquiry. They frequently include different combinations of technology development, question-driven inquiry, and exploration of phenomena.

The early development of the microscope, for example, revealed an invisible world that was then studied extensively in ways that led to entirely new understandings of life and of roles for technology in scientific investigation (Wilson, 1995). Hypothesis testing played only a minor role in this revolution, which involved exploratory investigation of regularities among microscopic phenomena. Again,

general questions and the exploration of new technologies, rather than strict hypotheses, have propelled the development of chemistry throughout its history (Caldin, 2002). In biology, despite the efforts of some commentators to reconstruct all of Charles Darwin's work as hypothesis testing (Ayala, 2009), it is clear that Darwin spent over two decades identifying regularities in several biological domains and codifying the phenomena to be explained in the *Origin* (Hodge, 1983). Similarly, in contemporary molecular biology, the accumulation of high-throughput data and technology-driven developments point to the importance of a diversity of non-hypothesis-driven practices (e.g., Brent, 2000; Aebersold et al., 2000; Franklin, 2005).

This broader range of practices involves open-ended and nonlinear approaches to specified—but sometimes vague and open—questions, problems, and ranges of phenomena. Inquiries tend to ramify as successive investigations produce partial answers that modify the original aim of inquiry, tighten its focus, and give rise to additional lines of research. Although such practices are broadly exploratory in nature, without pre-ordained endpoints, they are not necessarily preliminary or unfocused. Question-driven inquiry is often highly rigorous, and it occurs throughout many stages of investigation, often yielding an expansion of the domain of inquiry (Glass, 2006). Exploratory work frequently involves systematic variation of precisely defined parameters. It commonly focuses on ascertaining regularities or on individuating and characterizing previously unknown or neglected entities and processes (Steinle, 1997; Franklin, 2005). Technology-oriented modes of investigation encompass the development of entirely new instruments and techniques, the modification or application of old ones in new contexts, and the transformation of biological phenomena into new technologies. Frequently, these modes of scientific practice combine with the reconfiguration or replacement of conceptual frameworks. The discoveries that microRNAs contribute to eukaryotic gene regulation and that proteorhodopsin is present in marine bacteria serve as illuminating examples of such conceptual shifts and of how various modes of practice work in combination over time.

Unlike hypothesis-driven science, these broader practices are often not discipline bound, due to their inherent tendencies to step outside the boundaries of current technologies, theories, and community alliances. But despite the major differences between hypothesis-driven and broader modes of investigation, they are not dichotomous or mutually exclusive categories of practice. Hypothesis testing can be conceived of as specific and refined questioning that works best in appropriately narrowed contexts of inquiry. Typically, question-driven or exploratory work is required to generate the detailed understanding and background knowledge associated with such contexts. Most scientific inquiry will, therefore, consist of a range of investigative modes, including hypothesis testing, in repeated interplay with one another. This iterative interaction is arguably the engine of science and the characteristic that marks it as a special form of human inquiry.

Funding Agency Guidelines

Biologists may not need sophisticated philosophical accounts of the connections between these practices in order to do science successfully. But it is important to understand how these modes of investigation and their dynamic relationships are reflected in the philosophies of scientific practice that guide funding agencies (Wilkins, 2001). Frank Gannon recently published a parody of peer review in which two referees urge rejection of a fictional funding application by Charles Darwin. They criticize the proposed research for lacking an analysis of societal impact and also express concern about “the lack of a clear hypothesis to guide the enormous research task you propose: you could end up wasting your time and our money in an open-ended fishing exercise” (Gannon, 2009).

Amusing as it is to think of Darwin being turned down for the research program that underpinned his development of the principle of natural selection, it is worth reflecting on the fact that the criteria currently used by funding agencies would likely require rejection of a contemporary (molecular) equivalent of his proposal due to its exploratory and discovery-driven nature.

Although numerous funding agencies offer publicly funded support to life sciences research, we have focused on two US bodies, the NIH and NSF, and one UK research council, the BBSRC. These choices of agency allow us to make a rough comparative evaluation of philosophies of funding, not just cross-nationally but also between agencies with different research briefs, ranging from more strictly biological (BBSRC), to life sciences in general including biomedical investigation and clinical applications (NIH), to an even broader mélange of disciplines and themes, reaching well beyond biology (NSF).

The NIH offers very clear statements about what hypothesis testing is and why it is so central to research and the evaluation of funding applications (http://grants.nih.gov/grants/writing_application.htm). In its general advice to applicants on writing standard grant proposals, the agency tells inexperienced applicants that “A strong grant application is driven by a strong, solid hypothesis with clear research objectives. The specific aims are a formal statement of the objectives and milestones of the research project towards testing the hypothesis.... At its core, the typical, successful R01 grant application has a testable hypothesis in an area of high interest.”

These sentiments are expressed even more directly by the National Institute of Allergy and Infectious Disease, in the section entitled, “Create a solid hypothesis.” There, the agency instructs applicants to construct the entire proposal around a seemingly simple hypothesis (<http://www.niaid.nih.gov/ncn/grants/cycle/part02.htm#e>). “Most top-notch NIH grant applications are driven by strong hypotheses. Generally applications should ask questions that prove or disprove a hypothesis rather than search for a problem or simply collect information. Think of your hypothesis as the foundation of your application—the conceptual underpinning on which the entire structure rests. Your experimental results will prove or disprove your hypothesis.”

The situation, however, is more nuanced than these bald statements might indicate. To some degree, the NIH also recognizes the value of exploratory research. Its Exploratory/Developmental Research Program (R21) features a dif-

ferent application mechanism, providing a broader means by which problems of biological knowledge or clinical practice can be addressed. Emphasis is placed on novelty, discovery, and innovativeness: "The evolution and vitality of the biomedical sciences require a constant infusion of new ideas, techniques, and points of view. These may differ substantially from current thinking or practice and may not yet be supported by substantial preliminary data" (<http://grants.nih.gov/grants/guide/pa-files/PA-06-181.html#PartII>). The NIH supports various forms of innovative research through a range of "Funding Opportunities," some announced very recently (e.g., <http://grants.nih.gov/grants/guide/rfa-files/RFA-OD-09-004.html>). Thus, the NIH review criteria do not identify strict hypothesis testing as necessary or desirable for all types of funding.

The NSF follows a similar strategy. Hypothesis testing is taken as the norm, the prescribed mode of scientific practice: "Hypothesis testing: The standard model of the classical approach to scientific research in which a hypothesis is formulated before the experiment to test its truth" (Glossary in <http://www.nsf.gov/pubs/2002/nsf02057/>). But despite subscribing to this philosophy of science, NSF review guidelines do not mention hypotheses. Reviewers are merely instructed to assess proposals for "intellectual merit" and "broader impact" (https://www.fastlane.nsf.gov/NSFHelp/flashhelp/fastlane/FastLane_Help/fastlane_help.htm#welcome_to_the_fastlane_help_system.htm; scroll down to Proposal Review). The NSF also offers funding for exploratory research, previously under the banner of "Small Grants for Exploratory Research" (SGER), and now renamed "EARly-concept Grants for Exploratory Research," with the appealing acronym of "EAGER" (http://www.nsf.gov/pubs/policydocs/pappguide/nsf08_1/gpg_2.jsp#IID1). EAGER and SGER are mechanisms to fund "preliminary work on untested and novel ideas" and are designed as supplements to standard research routes (http://www.nsf.gov/funding/pgm_summ.jsp?pims_id=501093).

The BBSRC is the primary patron of life science research in the UK. It takes a less prescriptive view of how research should be done but is more directive

than the NIH and NSF about the framing of research questions, requiring that they be asked within particular fields of inquiry (e.g., systems biology, synthetic biology, aging research). Only very occasionally is hypothesis-driven research mentioned as the norm of scientific practice. The principal example we have found is in a mock referee's report that accompanies the guidelines for referees. In the section on "Scientific excellence," referees are requested to comment on the "clarity of hypotheses or aims and objectives" in relation to the strengths and weaknesses of the experimental design (http://www.bbsrc.ac.uk/funding/referees/referee_report_example.pdf). The referee guidelines themselves, however, do not specify the need for clear hypotheses. In part, this open approach to practice may be because of the BBSRC's present emphasis on systems biology. The systems biology section of the webpage devoted to "Our science" represents the agency's conceptualization of scientific practice as one of iterativity between hypothesis generation and testing (http://www.bbsrc.ac.uk/science/systems_approach.html).

Evaluating Funding Agency Philosophies

Although it is encouraging to see increasing recognition of broader strategies in research practice, there are several ways in which the structure of these funding programs does not adequately reflect the scientific importance and pervasiveness of non-hypothesis-driven modes of investigation. First, the NIH's R21 program defines "exploratory" projects as those that "assess the feasibility of a novel area of investigation or a new experimental system" or that could lead to the development of "novel techniques, agents, methodologies, models or applications." The other innovative award programs are even more explicit about focusing on new approaches or "out-of-the-box" ideas. The problem with these definitions is that they treat broader non-hypothesis-driven activities as preliminary, rather than as profoundly important for *ongoing* scientific activity. By separating exploratory grants from standard ones and limiting them to the early stages of research, this funding policy may hamper researchers from

moving back and forth between different approaches in advanced research projects. A further limitation is that the funding for R21 awards is capped at \$275,000 over a 2 year period, whereas standard hypothesis-driven R01 awards are not limited and can cover a 5 year time frame.

At the NSF, the same assumption that exploratory work is preliminary, rather than encompassing or interwoven with hypothesis-driven research, pervades the SGER and EAGER guidelines. Non-EAGER research is, in fact, described as "regular" research. Moreover, as in the case of the NIH, the funding for these awards is limited. In the first 4 weeks of 2009, for example, SGER Awards constituted just 16 out of 555 total NSF awards. Of the approximately 2000 SGER grants awarded between 1995 and 2008, only about 20% went to projects of a biological nature (<http://www.nsf.gov/awardsearch/>). Furthermore, a 2005 NSF Background Paper discusses numerous limitations of current programs for funding transformative, innovative, and exploratory research. Among the serious obstacles to the success of such funding programs, it specifically identifies poor definitions of this category of research, the inability to measure innovation and the success of exploratory investigations, and the conservatism of reviewers, especially when such programs are multidisciplinary (http://www.nsf.gov/nsb/committees/archive/tr/2005_08_bkgd_paper.pdf).

As already noted, the BBSRC does not explicitly separate hypothesis-driven research from broader approaches. What is emphasized is the interplay between hypothesis-generating and hypothesis-testing research strategies, especially in the context of systems biology (<http://blogs.bbsrc.ac.uk/index.php/tag/bbsrc/>; http://www.bbsrc.ac.uk/science/systems_approach.html). Although a range of practices appear to be implicit in this account, they are conceptually subordinated to devising, testing, and confirming hypotheses and models. In fact, the primary scientific output of any such hypothesis-focused schema would be hypotheses and models in and of themselves. We agree that hypothesis generation and testing are important to science at many points in a wider topography of inquiry but believe

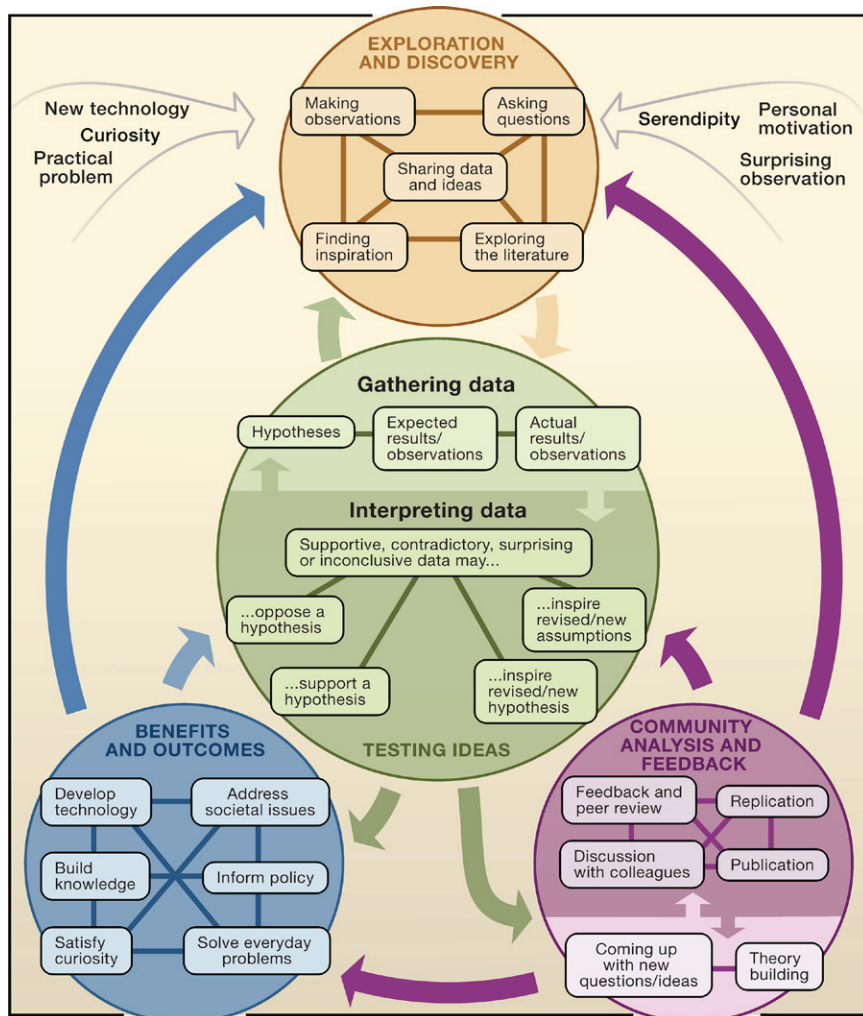


Figure 1. How Science Works

This depiction of science emphasizes the iterativity, interactivity, and fluidity of the scientific process. At its core are activities that involve testing ideas with data. This core is connected to an encompassing matrix of exploration and discovery, community analysis and feedback, and benefits and outcomes. All three outer processes feed into and receive feedback from the core process of testing ideas. Modified from Understanding Science, University of California Museum of Paleontology (http://undsci.berkeley.edu/flowchart_noninteractive.php).

that any practical philosophy of science has to recognize an overarching aim of producing understanding and effectively intervening in the world. The ways in which that aim is realized need to be decided pragmatically, not through idealizations about hypotheses, which too easily become overly simplistic proxies for good science and questionable means for identifying bad science.

A Broader Account of Scientific Practice

Is there a better representation of scientific processes available? An alternative is the four-sphere dynamic (Figure

1), produced with NSF funding by the "Understanding Science" project at the Museum of Paleontology, University of California at Berkeley, in collaboration with a diverse group of scientists and teachers.

Described by one commentator as "a dynamic alternative to the scientific method" (http://www.aibs.org/eye-on-education/eye_on_education_2009_01.html), the figure sets out a range of activities, aims, inputs, and outputs that include exploration, data gathering and analysis, hypothesis testing and revision, technology development, and community scrutiny. This four-sphere scheme

is a valuable correction of a narrow and exclusive representation of science as linear hypothesis testing, but we think it still does not capture sufficiently the deep and pervasive interplay between the distinct methodologies that constitute scientific inquiry. For example, exploration is placed primarily at the start of the scientific process and technological development at the end. We suggest instead that exploratory and technology-oriented research flow through all the other spheres of the diagram and enter iteratively into the core activities that are labeled in the diagram as "testing ideas." Although the authors of the diagram aim to overcome standard representations of hypothesis-driven science, the labels of these spheres—despite the two-way arrows between them and the other spheres—still fail to capture effectively the recurrence of exploratory, question-driven, and technology-oriented scientific investigation at every phase of scientific work.

A more complete picture of scientific activities (visual or verbal) would require a thorough investigation of the roles of different strategies in the overall practice of science. One approach that may help accomplish this is to examine key activities that characterize particular research contexts. These contexts are shaped by the background ontology of various disciplines (which may offer differing accounts of the nature of the phenomena and of underlying entities thought to explain those phenomena), claims about the reliability of available technology and modeling practices, the procedural assumptions of different disciplines (e.g., views about what constitutes sufficient and reliable evidence), and the culture of the scientific community. More open approaches thrive especially (but not exclusively) in a broad context of inquiry, where the precise nature of the phenomena is partially known but still contested, the instrumentation is developing in several directions, and the background knowledge is multiperspectival and multidisciplinary. The narrowing of such contextual information, so that a tightly focused hypothesis can be tested, occurs only in specific situations, often as a product of mono-disciplinary lines of investigation, and is likely to be centered in a short phase of inquiry.

For example, at first, molecular biology maintained a semblance of following the hypothesis-driven model. But when thousands of molecules involved in vast networks of activity are being studied, as they are in systems biology, a straightforward hypothesis-driven approach is insufficient to elucidate how major findings are made (Brent, 2000; Franklin, 2005). Linear modes of inquiry that narrow the research question so that it can produce a yes/no or true/false answer are then unlikely to capture the phenomena or characterize scientific practice adequately. The four-sphere diagram clearly recognizes this, but by retaining the notion of “testing ideas” as the core of the scientific process, it is likely to perpetuate a narrower view of science than is realistic.

We are certainly not advocating that all approaches and methodologies should be seen as equal. Scientific activity is, in part, the process of sorting out which approaches and accounts are valuable and which ones offer insufficient or misguided understanding. These judgments are contextually limited, but context, as we have argued, is most often multidimensional. We *do* think that no single scientific method applies in all cases, and that it is an open question as to which modes of investigation work best under particular circumstances.

Conclusions

Better representations of how science works will be useful for the public understanding of science, scientists' conception of science, and the appreciation by funding agencies of good science. Developing a multidimensional account of scientific inquiry that does justice to a broader conception of investigative strategies and their interplay is not the task of a few people with metascientific interests, but one that properly belongs to the communities it will affect. Recent molecular biology research showcases the difficulty of confining investigations to the contexts set by a single discipline or to one category of investigative tools. Even without a full picture of how science works in such complex settings, we suggest that more attention to the interplay between activities that include exploration, question-driven inquiry, and technology development, as well as hypothesis-driven investigation, will improve funding agency guidelines, expectations, and the norms of practice that they are enforcing and reinforcing.

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